

HAT-P-36b

R-0962

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_230204 · created 2026-07-29T16:12:26+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459979.9349 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.13 +/- 0.013
Transit depth (Rp/Rs)^2	1.7 +/- 0.33 [%]
Orbital Inclination (inc)	87.12 +/- 1.83
Airmass coefficient 1 (a1)	0.9498 +/- 0.008
Airmass coefficient 2 (a2)	0.043 +/- 0.049
Scatter in the residuals of the lightcurve fit is	0.84 %
Best Comparison Star	#3 - [525, 139]
Optimal Aperture	3.49
Optimal Annulus	14.38
Transit Duration (day)	0.0953 +/- 0.0027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.13 ± 0.013 vs 0.126 (deviation 0.27σ)
Duration fit vs published	0.0953 d vs 0.0925 d (deviation 0.92σ)
Mid-time O–C	-3.6 min
Depth SNR	8.9
Residual scatter	0.84 %

Light curve

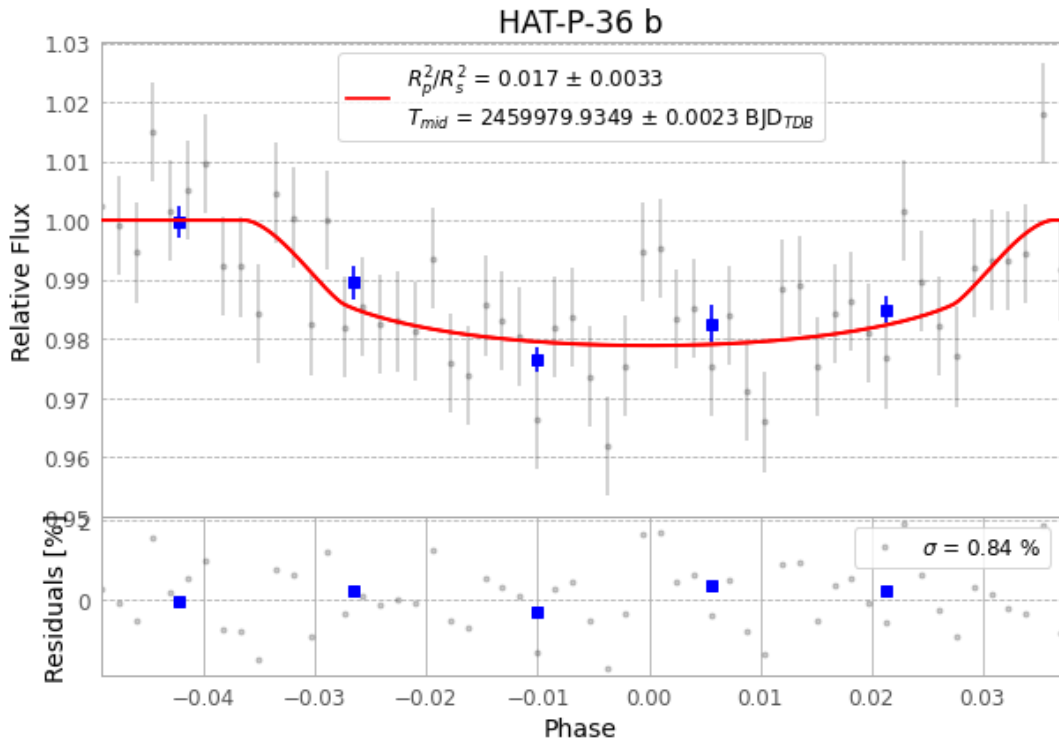


Figure 1 — FinalLightCurve_HAT-P-36 b_2023-02-04.png (SHA-256 abd00d24e70ad91f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [532, 155], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4a815104ec60a198...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

56 FITS frames (37 MB), HATP-36230204084515.FITS → HATP-36230204113015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-230204.log (3579b1267565...), FinalLightCurve_HAT-P-36 b_2023-02-04.png (abd00d24e70a...), FinalLightCurve_HAT-P-36 b_2023-02-04.csv (21b691e32721...)

Compute resources

Wall time 175.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 16:12Z.